

To Do *Space*

Activity, class, sequence
diagrams
And Non-Functional
requirements

Tasktronomy
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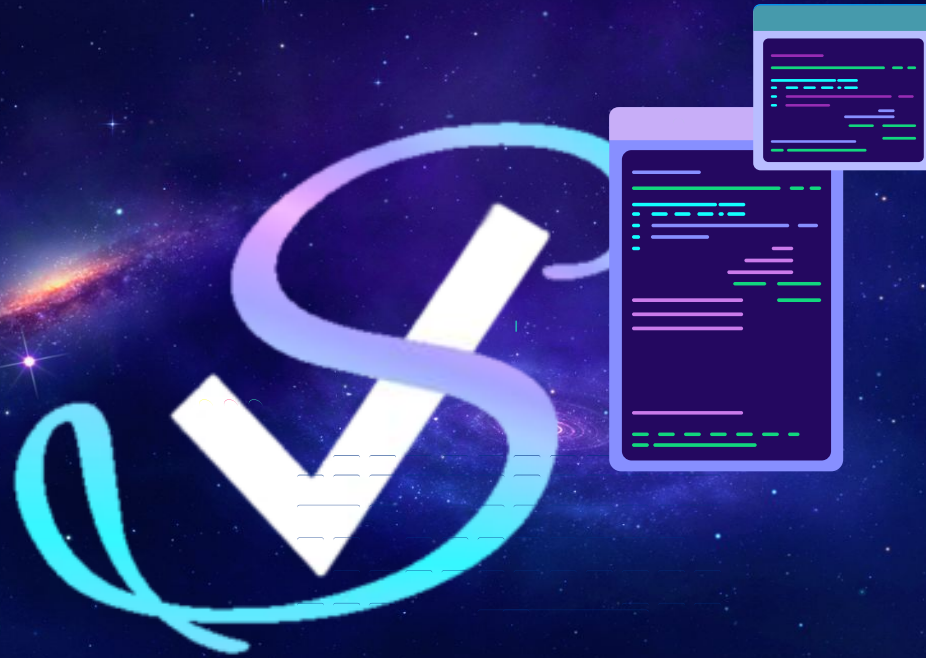


TABLE OF CONTENTS

01

Main application
flows

02

Class example:
Task creation

03

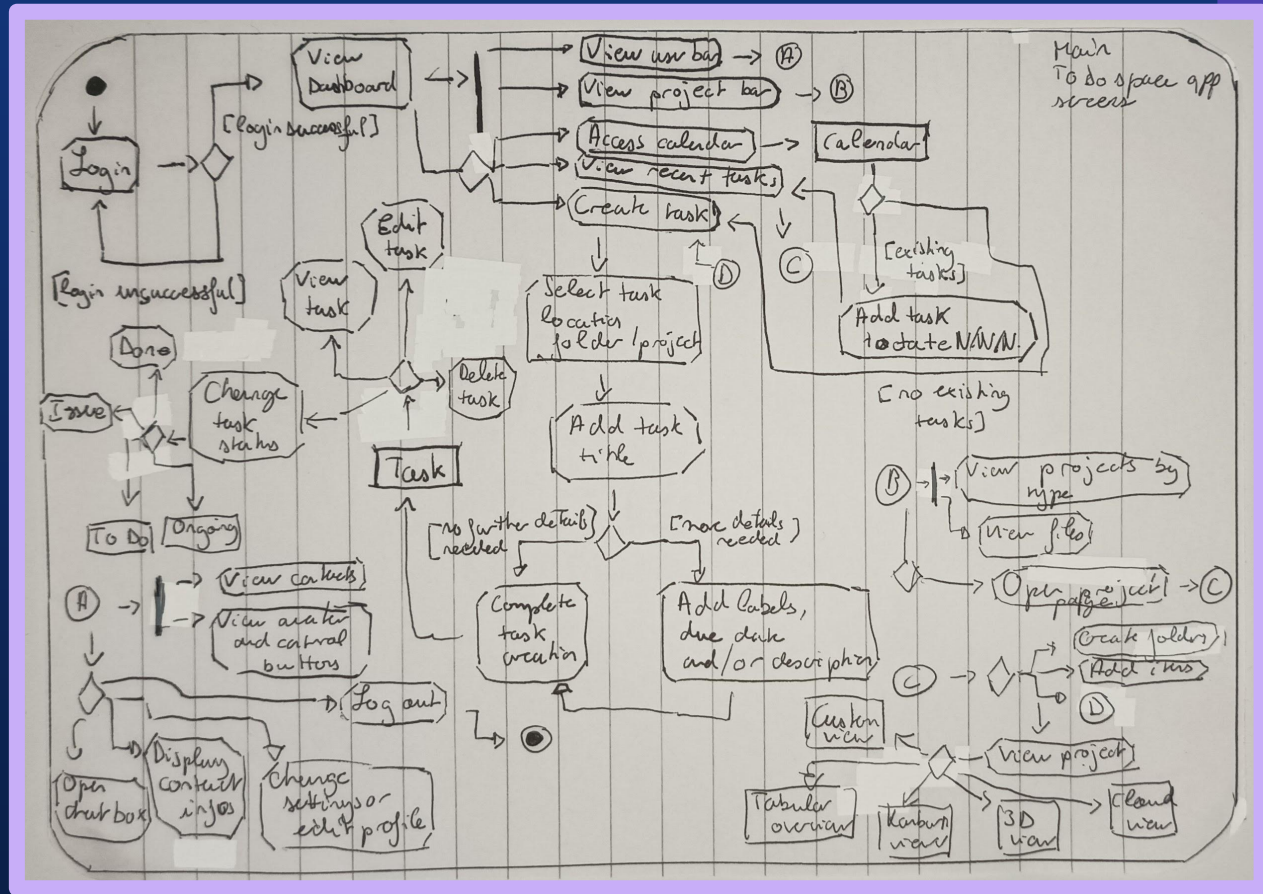
Sequence diagram
example: Task creation

04

Non - Functional
requirements

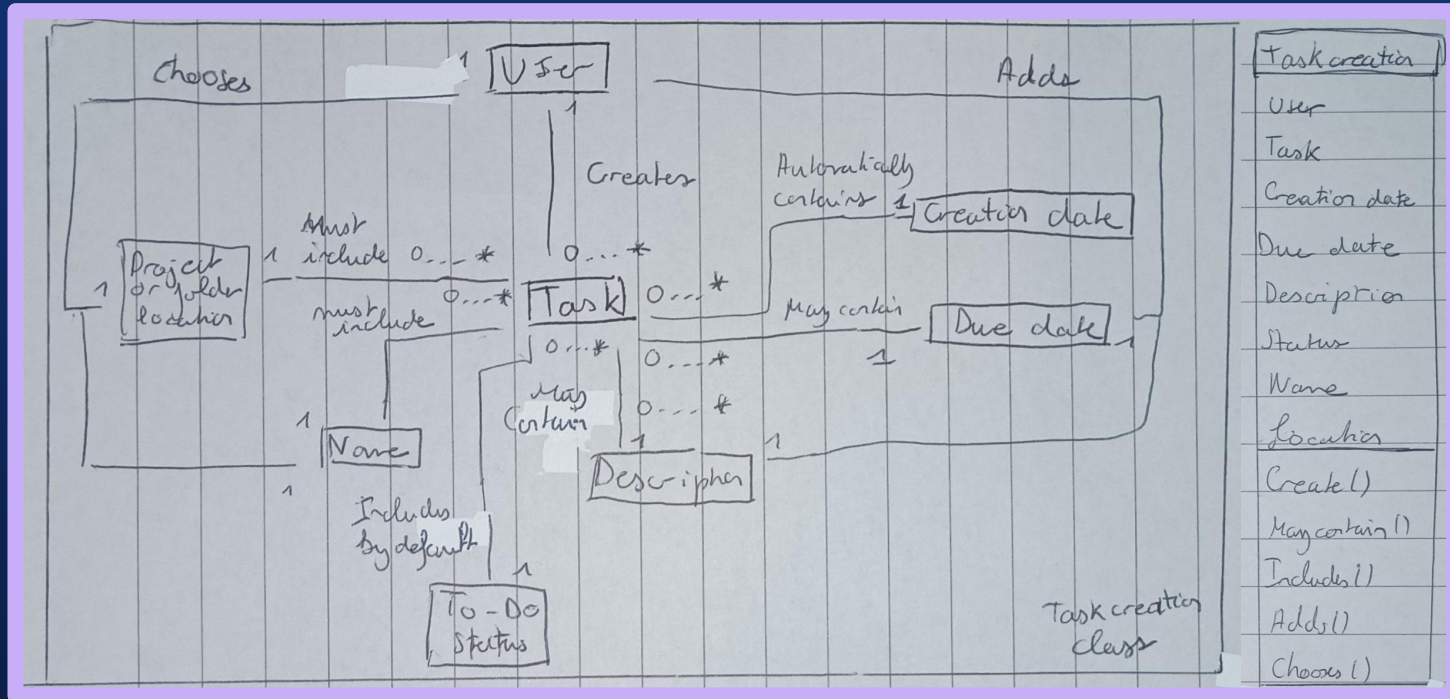


Activity diagram of the main features:



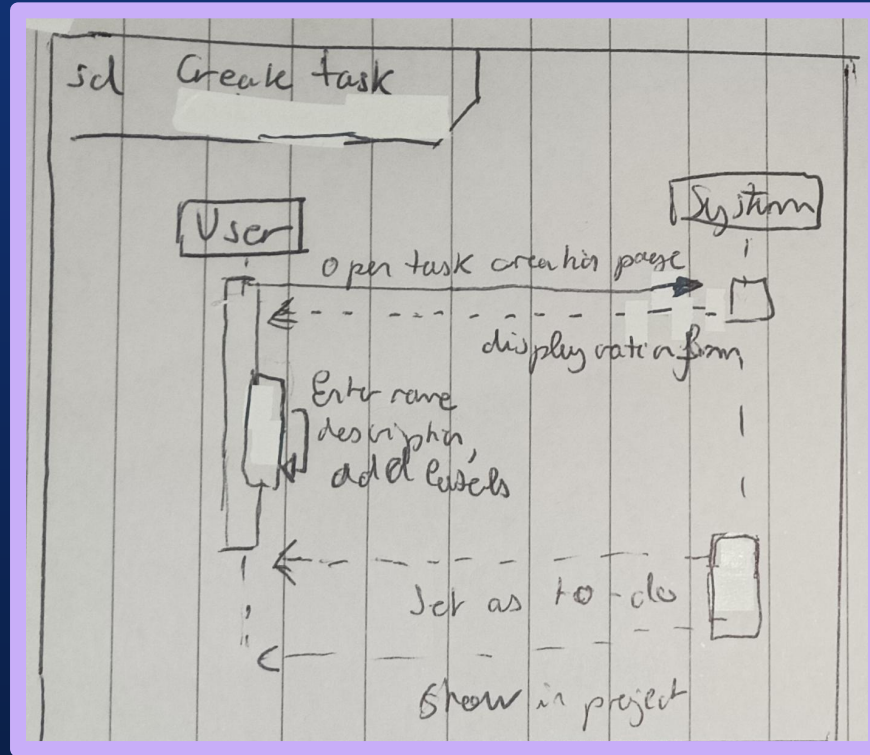
Task creation class:

As an example of the overall application's classes, here is the UML class diagram of the task creation feature: one of the core components of the application.



Sequence diagram example: Task creation

03





Non functional requirements:

Security requirements:

- **Password strength:** > 12 characters, A & a, > 1 special character, < 4 digits in a row
- **2 factor authentication:** Disabled by default but available, with SMS/email/Google authenticator
- **Authentication attempts track & captcha puzzle:** The system records all authentication attempts. If > 10 attempts are made, the login is blocked. Captcha puzzles also appear once in a while
- **Encryption:** All data sent from and to the server gets encrypted, preferably through 3DES and SHA512 for password hashes
- **Access and control:** all data access is denied to third party apps except if they are on the authorized apps list



Non functional requirements:

04



Interoperability requirements:

Format: Application based on HTML to be viewed on any browser





Non functional requirements:

Performance requirements:

Optimal latency: <35ms

Latency that is considered reasonable: < 300ms

Optimal throughput: >10 mb/s

Throughput that is considered reasonable: < 100 kb/s

Switch to offline mode: Latency > 300ms, throughput > 100 kb/s

Query processing time (read only): 1 second

Query processing time (write): 2 seconds

Session duration: expect an average 20 minutes active, 4 hours in the background

Session expiration: Set by user, 2 hours of inactivity by default





Non functional requirements:

Availability requirements:

Availability requirements for different geographies: Not available in countries that do not support the linked applications (ex. Google in china)

Frequency of scheduled planned outage activity that is acceptable: 2-3 hours every 3 months

Expected unplanned outage activity: 25-30 hours during the launch year

Expected overall expected server uptime: 99.995%

Service Desk Availability (for non-critical situations): Q&As open 24/7, support available weekly

Support team availability (for critical situations): Daily





Non functional requirements:

Scalability requirements:

Ability to scale up to n active users in a year: To be determined, expect 10 000 users at launch

Number of concurrent users during peak periods: TBD, expect 500 users at launch

Session duration: expect an average 20 minutes active, 4 hours in the background

Query processing time (read only, open page): 1 second

Query processing time (write): 2 seconds

Number of tasks per free user: 1000

Data quantity stored per free user: 5GB

